Starburst

Ibis Data Pipelines
Starburst Workshops

A hands-on webinar
with Lester Martin from DevRel

Connection before content

Lester Martin – https://linktr.ee/lestermartin

- Developer Relations @ Starburst
  - Blogging & forums
  - Webinars & videos
  - User groups & events
  - Training & tutorials
- 30+ years of technology experience
  - Started journey on TRS-80 Model III
  - Played most roles, but a programmer at my core
  - ½ career in OLTP and ½ in data analytics
  - Decade+ of "big data" experience to include
    - Trino/Starburst, Hadoop, Hive, Spark
    - NiFi, Kafka, Storm, Flink
    - HBase, MongoDB

devrel@starburst.io

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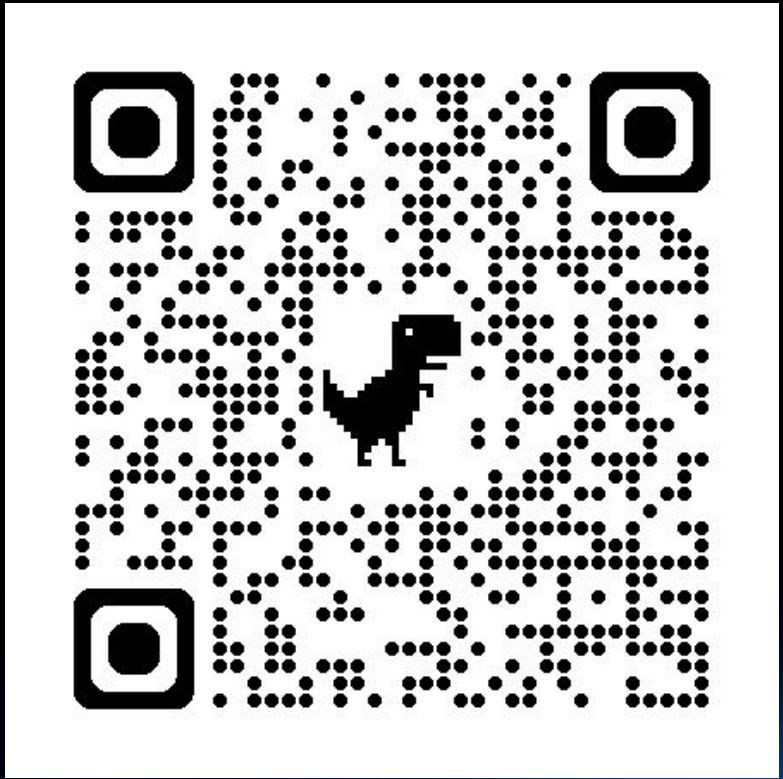
Environment setup

Setup and instructions located on Github

https://github.com/starburstdata/devrel/blob/main/workshops/ibis-pipelines/INSTRUCTIONS.md



Environment

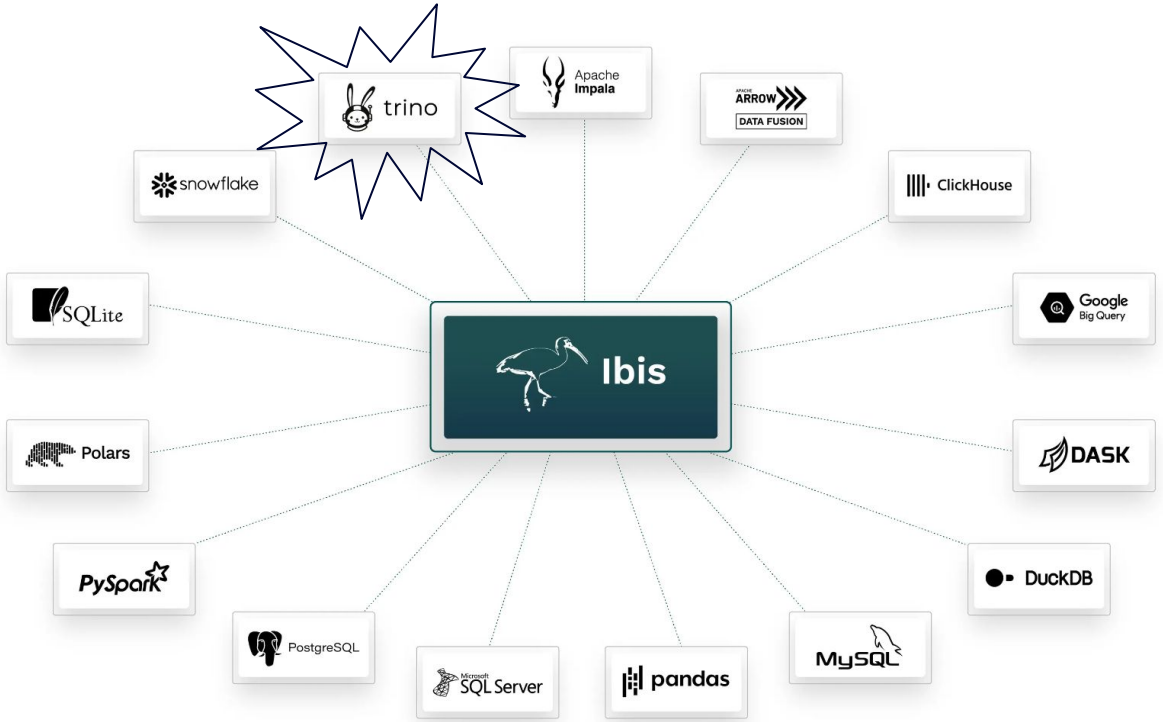
Using Starburst Galaxy and S3

Exercises

Jupyter notebook on GitHub

Ibis & Starburst (via Trino backend)

A common Dataframe API for data manipulation in Python, and compiling that API into a variety of backend native languages



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Hands-on

Validate env setup

Data pipelines in Python w/Ibis

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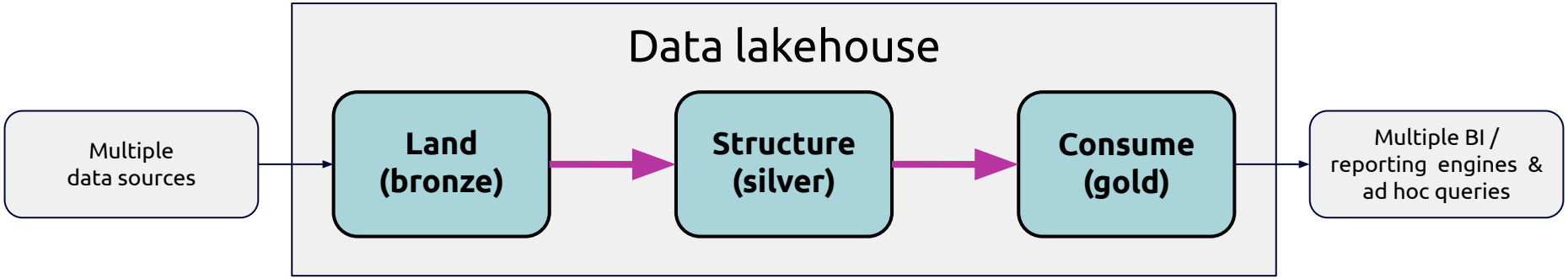
Activities across the medallion architecture



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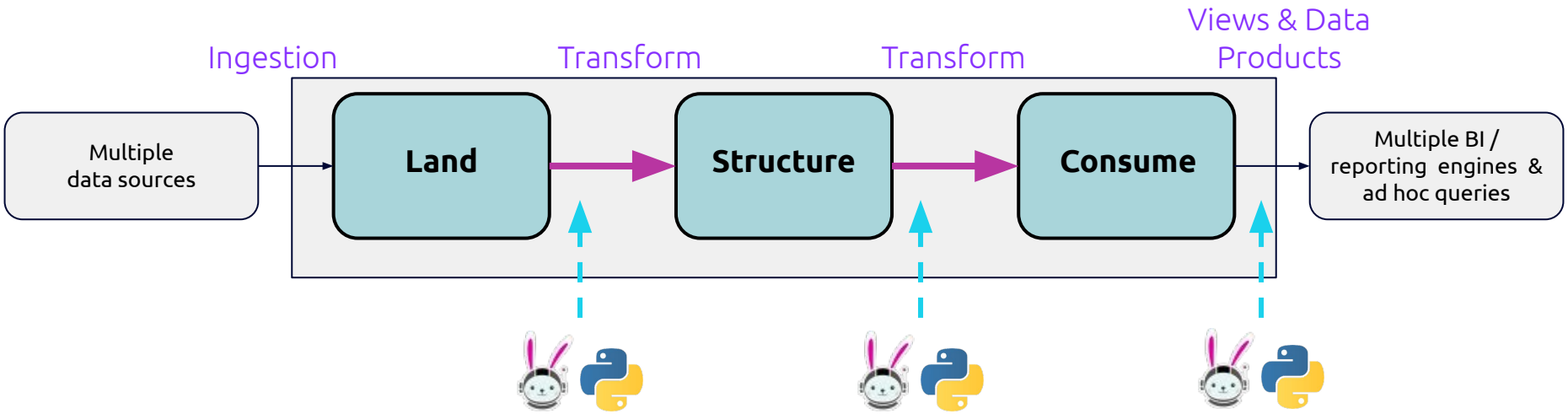
Data pipelines with SQL and/or Python



Data pipeline

Extract	*Load*	*Transform*	*Transform*	*Transform*

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SQL for transformation processing

```sql
SELECT
  CASE
    WHEN product_id IN (1, 2, 3, 4) THEN 'product_one'
    ELSE 'product_two'
  END as product,
  SUM(order_amount) as sales
FROM
  tmp_sales
GROUP BY
  CASE
    WHEN product_id IN (1, 2, 3, 4) THEN 'product_one'
    ELSE 'product_two'
  END
```

Pros

- Compact
- Easy to understand for simple logic

Cons

- No real time debugging
- Can’t perform other actions on results
- No unit tests
- Version Diffs likely hard to read

Complexity makes SQL even more difficult

```sql
SELECT
  order_month,
  order_day,
  COUNT(DISTINCT order_id) AS num_orders,
  COUNT(book_id) AS num_books,
  SUM(price) AS total_price,
  SUM(COUNT(book_id)) OVER (
    PARTITION BY
      order_month
    ORDER BY
      order_day
  ) AS running_total_num_books
FROM
  (
    SELECT
      DATE_FORMAT(co.order_date, '%Y-%m') AS order_month,
      DATE_FORMAT(co.order_date, '%Y-%m-%d') AS order_day,
      co.order_id,
      ol.book_id,
      ol.price
    FROM
      cust_order co
      INNER JOIN order_line ol ON co.order_id = ol.order_id
  ) sub
GROUP BY
  order_month,
  order_day
ORDER BY
  order_day ASC;
```

As complexity grows

- Schema changes are hard to debug
- Logic errors are difficult to find
- Collaboration and sharing is more difficult

Python Dataframes

A Dataframe is a Python data structure that represents a Table that can be operated on

PySpark Example

```python
from pyspark.sql import DataFrame, SparkSession
import spark.sql.functions as F

def read_sales_data(uri: str = 's3a://sales-data/customer-orders/2021/*') -> DataFrame:
    df = spark.read.parquet(uri)
    return df

def define_product(input_df: DataFrame) -> DataFrame:
    output_df = input_df.withColumn('product',
                                    F.when(
                                           F.col('product_id').isin([1,2,3,4]), F.lit('product_one')).otherwise(F.lit('product_two')
                                    )
                )
    return output_df

def agg_sales_by_product(input_df: DataFrame, gb: str = 'product', ag: str = 'order_amount') -> DataFrame:
     output_df = input_df.groupBy(gb).agg(F.sum(F.col(ag).alias('sales'))
     return output_df


df = read_sales_data()
products = define_product(df)
metrics = agg_sales_by_product(products)
```

Pros

- Reusable functions
- Version diffs are easy to read
- Easily create unit tests
- Could combine Python libraries like Numpy, Scipy
- Perform real-time debugging
- Easy to follow complex logic
- Allows for logging

Cons

- A bit verbose

Behind the scenes (1)

SQL Query
Human

Query text

```sql
SELECT
  c.first_name,
  c.last_name,
  c.estimated_income,
  a.products,
  a.cc_number,
  a.mortgage_id,
  cp.customer_segment,
  pp.cc_type
FROM
  glue.burst_bank.customer c
  JOIN mysql.burst_bank.account a on c.custkey = a.custkey
  JOIN mysql.burst_bank.product_profile pp on a.custkey = pp.custkey
  JOIN postgresql.burst_bank.customer_profile cp on c.custkey = cp.custkey
WHERE
  cp.customer_segment = 'diamond'
  and c.estimated_income > 200000
  and pp.cc_type = 'basic'
LIMIT 10
```

SQL Query
Dataframe

Query text

```sql
SELECT
  "first_name"
, "last_name"
, "estimated_income"
, "products"
, "cc_number"
, "mortgage_id"
, "cc_type"
, "customer_segment"
FROM
  (
   SELECT *
   FROM
     (
      SELECT *
      FROM
        (
         SELECT *
         FROM
           (
            SELECT *
            FROM
              (
               SELECT *
               FROM
                 ((
                  SELECT
                    "l_jnri_custkey" "l_jnri_custkey"
                  , "first_name" "first_name"
                  , "last_name" "last_name"
                  , "estimated_income" "estimated_income"
                  , "r_mkez_custkey" "r_mkez_custkey"
                  , "products" "products"
                  , "cc_number" "cc_number"
                  , "mortgage_id" "mortgage_id"
                  , "custkey" "l_belm_custkey"
                  , "cc_type" "cc_type"
                  FROM
                    (
                     SELECT *
                     FROM
                       ((
                        SELECT
                          "l_jnri_custkey" "l_jnri_custkey"
                        , "first_name" "first_name"
                        , "last_name" "last_name"
                        , "estimated_income" "estimated_income"
                        , "r_mkez_custkey" "r_mkez_custkey"
                        , "products" "products"
                        , "cc_number" "cc_number"
                        , "mortgage_id" "mortgage_id"
                        FROM
                          (
                           SELECT *
                           FROM
                             ((
                              SELECT
                                "custkey" "l_jnri_custkey"
                              , "first_name" "first_name"
                              , "last_name" "last_name"
                              , "estimated_income" "estimated_income"
                              FROM
                                (
                                 SELECT
                                   "custkey"
                                 , "first_name"
                                 , "last_name"
                                 , "estimated_income"
                                 FROM
                                   (
                                    SELECT *
                                    FROM
                                      glue.burst_bank.customer
                                   )
                                )
                             ) INNER JOIN (
                              SELECT
                                "custkey" "r_mkez_custkey"
                              , "products" "products"
                              , "cc_number" "cc_number"
                              , "mortgage_id" "mortgage_id"
                              FROM
                                (
                                 SELECT
                                   "custkey"
                                 , "products"
                                 , "cc_number"
                                 , "mortgage_id"
                                 FROM
                                   (
                                    SELECT *
                                    FROM
                                      mysql.burst_bank.account
                                   )
                                )
                             ) ON ("r_mkez_custkey" = "l_jnri_custkey"))
                          )
                       ) INNER JOIN (
                          SELECT
                            "custkey" "custkey"
                          , "cc_type" "cc_type"
                          FROM
                            (
                             SELECT
                               "custkey"
                             , "cc_type"
                             FROM
                               (
                                SELECT *
                                FROM
                                  mysql.burst_bank.product_profile
                               )
                            )
                       ) ON ("custkey" = "l_jnri_custkey"))
                    )
                 ) INNER JOIN (
                    SELECT
                      "custkey" "r_wza9_custkey"
                    , "customer_segment" "customer_segment"
                    FROM
                      (
                       SELECT
                         "custkey"
                       , "customer_segment"
                       FROM
                         (
                          SELECT *
                          FROM
                            postgresql.burst_bank.customer_profile
                         )
                      )
                 ) ON ("r_wza9_custkey" = "l_jnri_custkey"))
              )
           WHERE ("customer_segment" = 'diamond')
        )
      WHERE ("estimated_income" > CAST(200000 AS BIGINT))
     )
   WHERE ("cc_type" = 'basic')
)
OFFSET 0 ROWS
LIMIT 10
```

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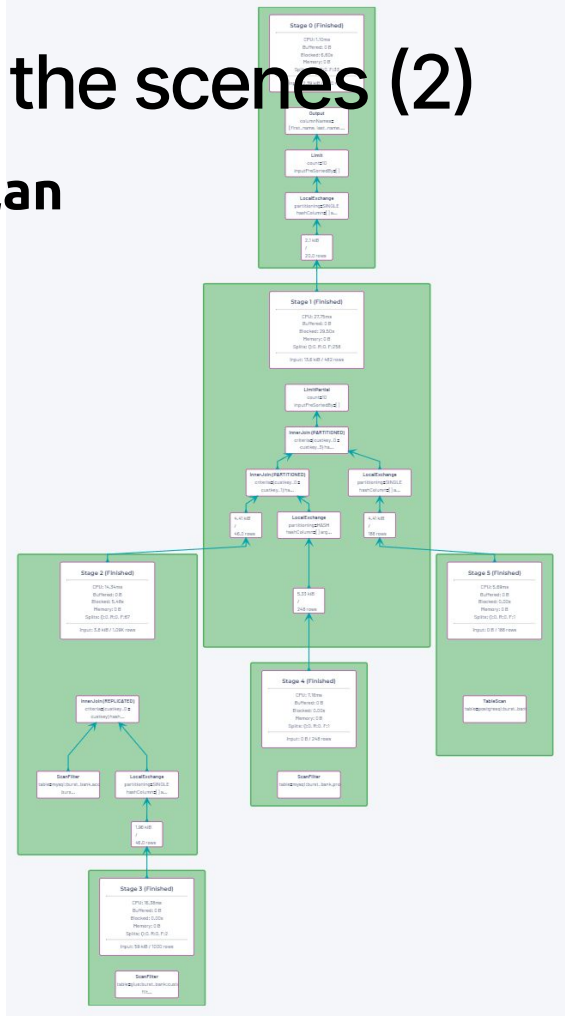
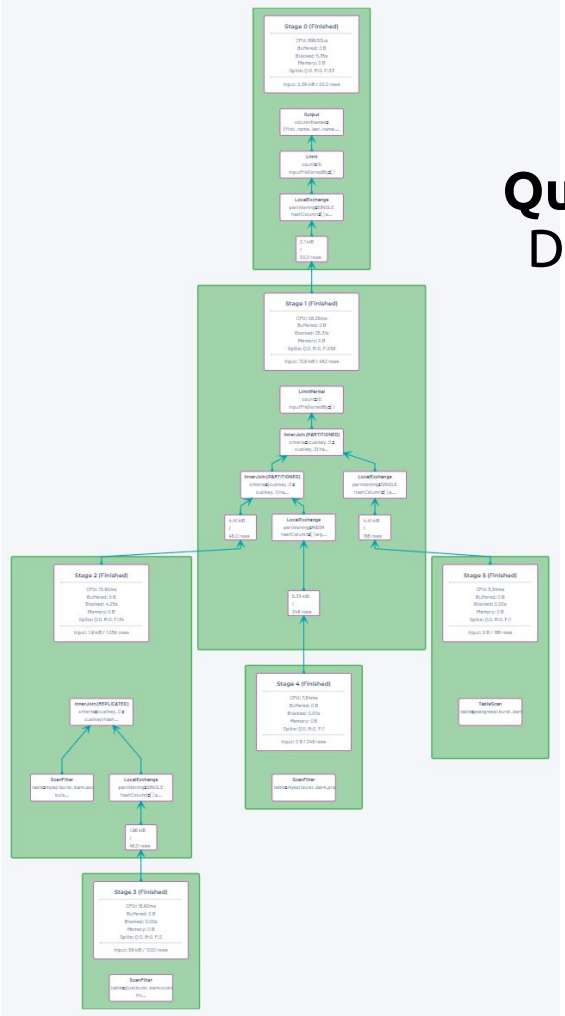
Behind the scenes (2)

Query Plan
Human



Query Plan
Dataframe



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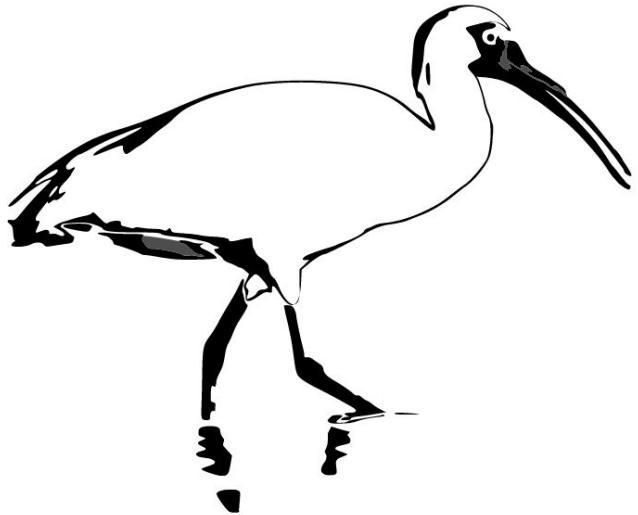
Dataframe API options with Starburst

PyStarburst

PyStarburst

Ibis



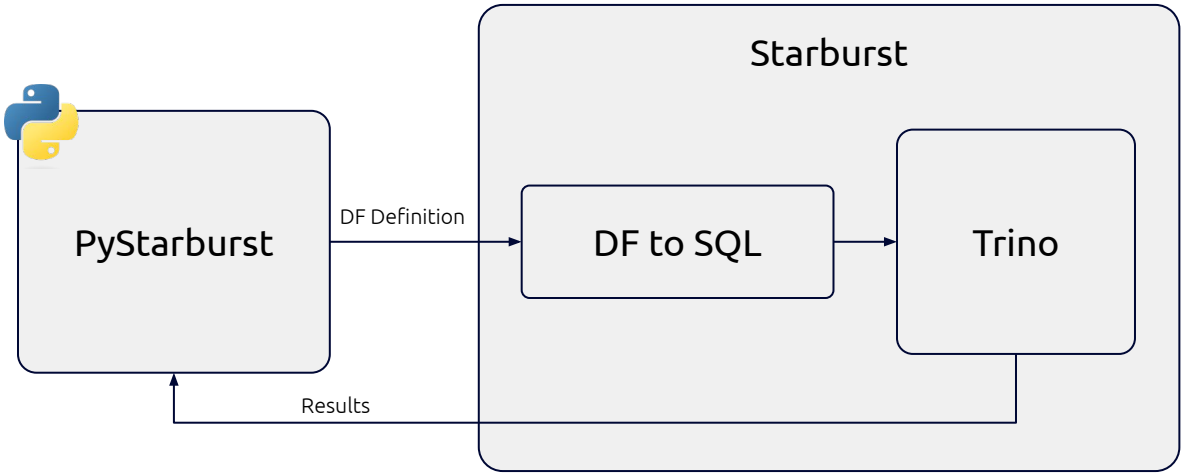
PyStarburst overview

```
df_missions = df_missions.with_column("date", f.sql_expr("COALESCE(TRY(date_parse(\"date\", '%a %b %d, %Y %H:%i UTC')), NULL)"))

print(df_missions.schema)

df_missions = df_missions\
    .filter(col("date") > datetime(2000, 1, 1))\
    .sort(col("date"), ascending=True)

df_missions.show()
```



- PySpark-like Syntax
- Lazy execution
- Python gets converted to SQL
- Heavy lifting done by Trino

Links: API Docs & Example Code

Only supported on Starburst Galaxy and Starburst Enterprise

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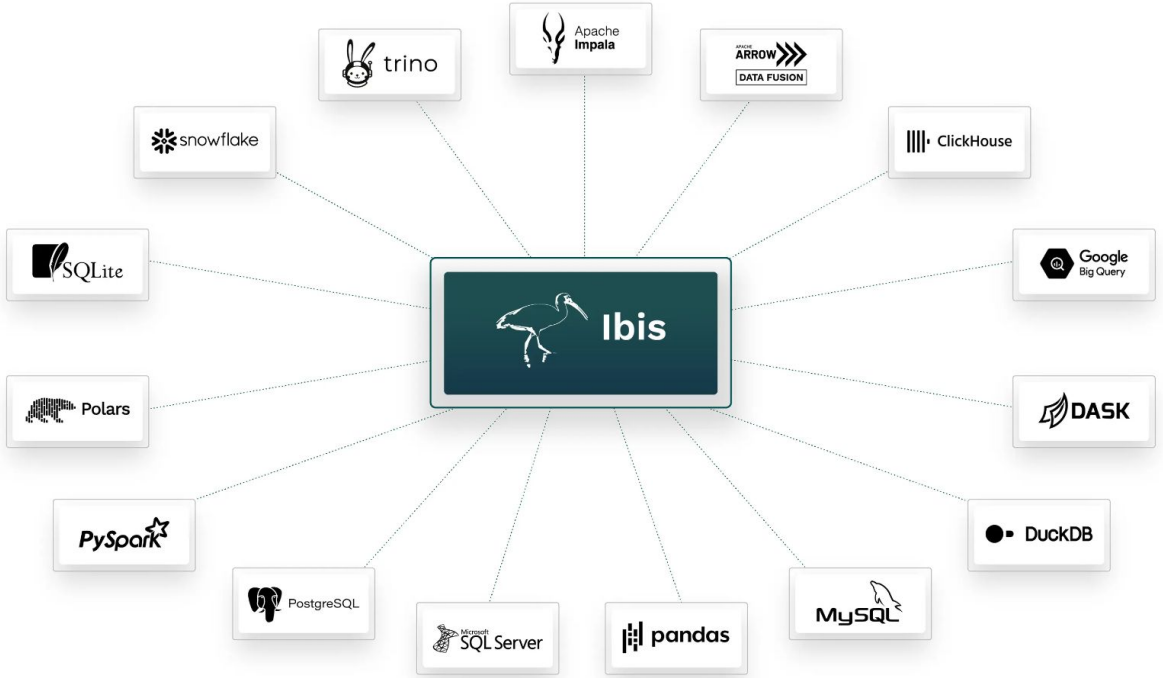
Ibis overview

A common Dataframe API for data manipulation in Python, and compiling that API into a variety of backend native languages



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Similar APIs

PyStarburst

```
nationDF = session.table("tpch.tiny.nation") \
            .drop("regionkey", "comment") \
            .rename("name", "nation_name") \
            .rename("nationkey", "n_nationkey")
```

```
custDF = session.table("tpch.tiny.customer") \
            .select("name", "acctbal", "nationkey") \
            .filter(col("acctbal") > 9900.0)

apiSQL = custDF.join(nationDF,
    col("nationkey") == nationDF.n_nationkey) \
            .drop("nationkey").drop("n_nationkey") \
            .sort(col("acctbal"), ascending=False)

apiSQL.show()
```

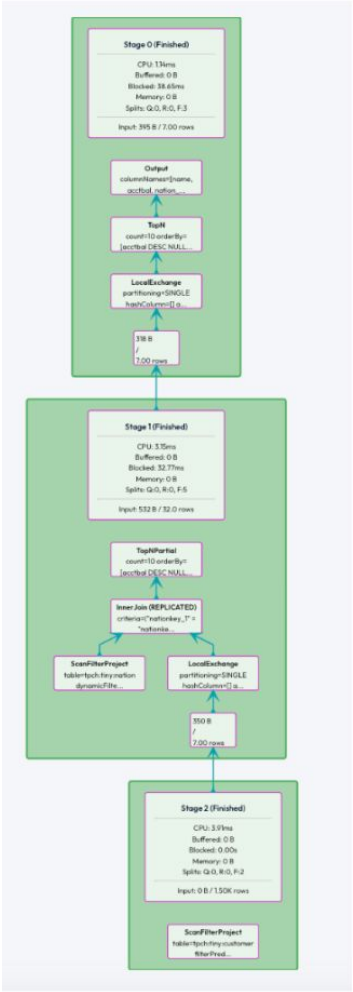
Ibis

```
nationDF = con.table("nation") \
            .drop("regionkey", "comment") \
            .rename(
                dict(
                    nation_name="name",
                    n_nationkey="nationkey"
                )
            )
```

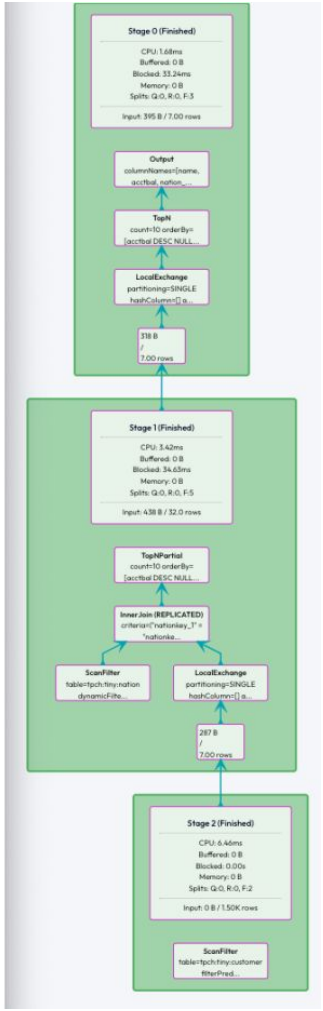
```
custDF = con.table("customer") \
            .select("name", "acctbal", "nationkey") \
            .filter(projectedDF["acctbal"] > 9900.0)

apiSQL = custDF.join(nationDF,
    custDF.nationkey == nationDF.n_nationkey) \
            .drop("nationkey", "n_nationkey") \
            .order_by([ibis.desc("acctbal")])

print(apiSQL.head(10))
```

Similar results

Query Plan
PyStarburst



Query Plan
Ibis



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Trino for your data pipelines

Trino is, and has been, an appropriate clustering technology for running you transformation processing jobs

- Fault-tolerant execution mode makes it even more robust
- Data engineers can leverage SQL and/or Python

trino

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Dataframe API recommendations

If you use Starburst products → Choose PyStarburst

- More similar to PySpark Dataframe API
- More likely to be optimized better for Trino than Ibis

If you are using OSS Trino → Choose Ibis

- PyStarburst is not supported here
- Gain optionality for backend execution engine

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Hands-on

Ibis in a notebook

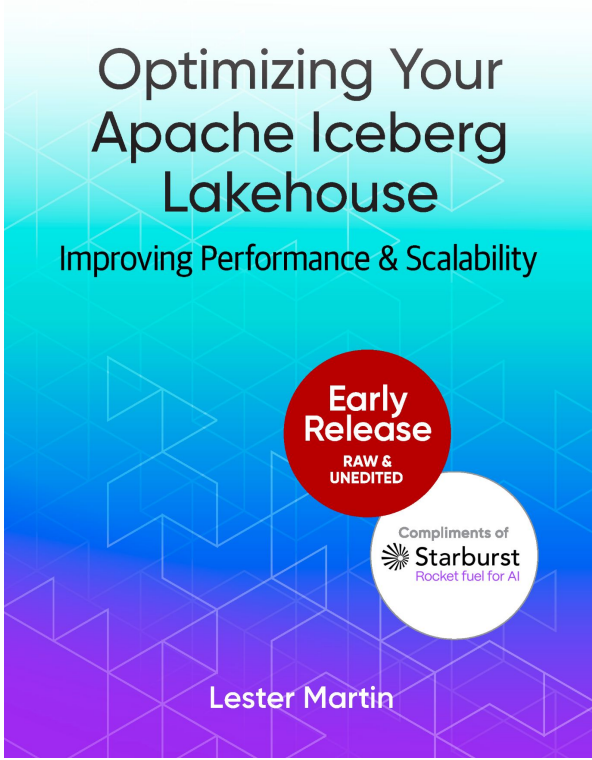
What are my next steps?

- https://devcenter.starburst.io/
- https://www.starburst.io/community/forum/
- https://www.starburst.io/learn/events-webinars/
- https://www.starburst.io/blog/pystarburst-the-dataframe-api/

A bit of shameless self-promotion…

Optimizing Your Apache Iceberg Lakehouse

Early Release #1



Demo artifacts at https://github.com/starburstdata/devrel/tree/main/workshops/ibis-pipelines

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Thank you!

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